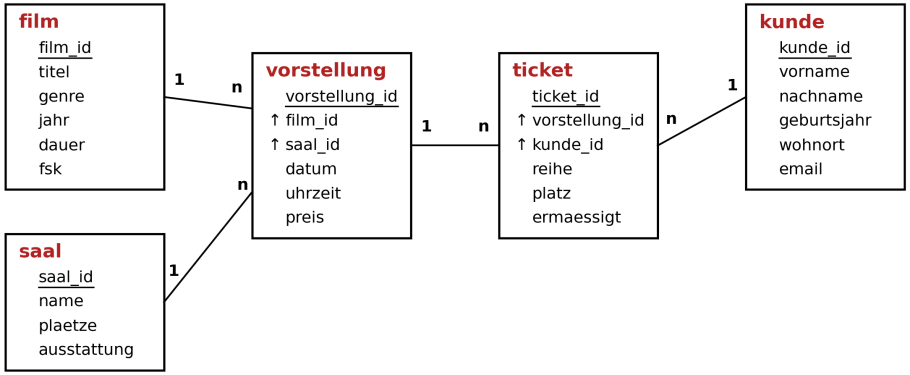
Merkhilfe SQL



Datenbank `kino.db`: Primärschlüssel unterstrichen, ↑ Fremdschlüssel, Linien mit 1 und n

Start

```
%LOAD kino.db r                                  Datenbank öffnen
SELECT name FROM sqlite_master WHERE type = 'table';
PRAGMA table_info(film);                         Spalten
```

eine Anweisung pro Zelle, ; am Ende

Reihenfolge der Teile

```
SELECT ... FROM ...
  JOIN ... ON ...
  WHERE ...
  GROUP BY ... HAVING ...
  ORDER BY ... LIMIT ...;
```

Spalten auswählen

```
SELECT * FROM film;
SELECT titel, jahr FROM film;
SELECT DISTINCT genre FROM film;          ohne Doppelte
SELECT dauer / 60.0 AS stunden            rechnen, umbenennen
ROUND(x, 2)                               runden
vorname || ' ' || nachname                Text verbinden
```

Zeilen auswählen mit WHERE

```
WHERE fsk = 12   <> != < > <= >=
WHERE genre = 'Krimi'                     Text in ' '
AND   OR   NOT   ( )
WHERE jahr BETWEEN 2000 AND 2009
WHERE genre IN ('Horror', 'Krimi')
WHERE titel LIKE 'Die %'                  % beliebig, _ ein Zeichen
WHERE email IS NULL   IS NOT NULL
```

Sortieren und begrenzen

```
ORDER BY dauer   ORDER BY dauer DESC
ORDER BY genre, titel                     mehrere Spalten
LIMIT 3   LIMIT 3 OFFSET 3
```

Aggregatfunktionen

```
COUNT(*)   COUNT(email)                   ohne NULL
COUNT(DISTINCT genre)
SUM(preis)   AVG(dauer)
MIN(jahr)   MAX(jahr)
ROUND(AVG(dauer), 1)
```

Gruppieren

```
SELECT genre, COUNT(*) AS anzahl
FROM film
GROUP BY genre
HAVING COUNT(*) > 2;                      Bedingung für Gruppen
```

Tabellen verbinden mit JOIN

```
SELECT f.titel, v.datum
FROM vorstellung AS v
JOIN film AS f ON v.film_id = f.film_id;
```

mehrere Tabellen:

```
FROM ticket AS t
JOIN vorstellung AS v
  ON t.vorstellung_id = v.vorstellung_id
JOIN film AS f ON v.film_id = f.film_id
```

JOIN und GROUP BY

```
SELECT f.titel, COUNT(*) AS tickets
FROM ticket AS t
JOIN vorstellung AS v ON ...
JOIN film AS f ON ...
GROUP BY f.film_id
ORDER BY tickets DESC;
```

LEFT JOIN

```
FROM film AS f
LEFT JOIN vorstellung AS v
  ON f.film_id = v.film_id                auch ohne Partner
WHERE v.vorstellung_id IS NULL            nur ohne Partner
COUNT(v.vorstellung_id)                   zählt auch 0
```

Unterabfragen

```
WHERE preis > (SELECT AVG(preis)
               FROM vorstellung)
WHERE film_id IN (SELECT film_id
                  FROM vorstellung)
WHERE kunde_id NOT IN (SELECT ...)
```

Funktionen

```
UPPER(t)   LOWER(t)   LENGTH(t)
SUBSTR(t, 1, 5)                           ab Position 1
REPLACE(t, 'alt', 'neu')
strftime('%d.%m.%Y', datum)               02.11.2026
strftime('%w', datum)                     '0' Sonntag
```

CASE

```
CASE WHEN fsk = 0 THEN 'alle'
     WHEN fsk < 16 THEN 'Jugend'
     ELSE 'Erwachsene'
END AS freigabe
```